Technical University of Crete
School of Electrical and Computer Engineering
Course: Optimization
Exercise 1 (80/1000)
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In this exercise, we will study simple concepts from Calculus of Several Variables (Taylor expansions), Convex Sets, and Convex Functions (Note: $R_+ = \{x \in \mathbb{R} | x \geq 0\}$, $R_{++} = \{x \in \mathbb{R} | x > 0\}$).

1: Let f: $\mathbb{R}^+ \to \mathbb{R}$ with f=$\frac{1}{1+x}$ Let $x_0 \in \mathbb{R}_+$ and define the first- and second-order Taylor approximations of f at x_0 as :

$$f_{(1)}(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$f_{(2)}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2$$

(a) Find the analytic expressions for functions f' and f''.
(b) Draw in a common plot f(x), $f_{(1)}(x)$ and $f_{(2)}(x)$ and, in order to understand the behavior of the approximations, consider various values of x_0.

Answer: The analytic expression for the function f' is actually the first derivative of function f.

$$f' = \frac{\partial f}{\partial x}$$

$$\frac{\partial f}{\partial x} = -\frac{1}{(1+x)^2}$$

The second derivative of f is computed as:

$$f'' = \frac{\partial f^2}{\partial^2 x}$$

$$\frac{\partial f^2}{\partial^2 x} = \frac{2}{(1+x)^3}$$

So, then the approximations are computed as

$$f_{(1)}(x) = \frac{1}{1+x_0} - \frac{1}{(1+x_0)^2} \cdot (x - x_0)$$

$$f_{(2)}(x) = \frac{1}{1+x_0} - \frac{1}{(1+x_0)^2} \cdot (x - x_0) + \frac{1}{(1+x_0)} \cdot (x - x_0)^2$$

$$f_{(2)}(x) = f_{(1)}(x) + \frac{1}{(1+x_0)} \cdot (x - x_0)^2$$

These were the formulas that were used in question (b) for the plots of the approximations in Matlab. In the second part of the problem, Matlab was used in order to design the approximations. The values of x_0 were defined from 1 to 9 and using a for-loop the graph of function f and its approximations were designed for the several values of x_0. The code is listed below and so are the plots. *Matlab Code:*

```
%% 1
syms x

% Define x values
x = linspace(0.1, 5, 1000);
f = @(x) 1./(1 + x);

% 1st and 2nd derivatives of f(x)
f_der1 = @(x) -1./((1 + x).^2);
f_der2 = @(x) 2./((1 + x).^3);

syms x0

% Define x0 values
x0_values1 = [2:5];

% we define different styles for lines
% to make the diagrams more clear
line_styles = {'-', '--', '-.'};

figure;
hold on;

for i = 1:length(x0_values1)
    x0 = x0_values1(i);
    f1 = f(x0) + f_der1(x0)*(x-x0);
    f2 = f(x0) + f_der1(x0)*(x-x0) + 0.5 * f_der2(x0) * (x-x0).^2;
    subplot(2,2,i);
    plot(x,f(x), line_styles{1}, 'LineWidth', 1, 'DisplayName',
    sprintf('f(x) for x_0=%d', x0));
    hold on;
    plot(x,f1, line_styles{2}, 'LineWidth', 1, 'DisplayName',
    sprintf('f1(x) for x_0=%d', x0));
    plot(x,f2, line_styles{3}, 'LineWidth', 1, 'DisplayName',
    sprintf('f2(x) for x_0=%d', x0));
    title(['x_0 = ' num2str(x0)]);
    xlabel('x');
    ylabel('y');
    legend('show');
```

```
    grid on;
    hold off;
end
```

In order to get a better visualization of the relationships between the function f and its Taylor approximations in relation with x_0, the command 'subplot' was used.

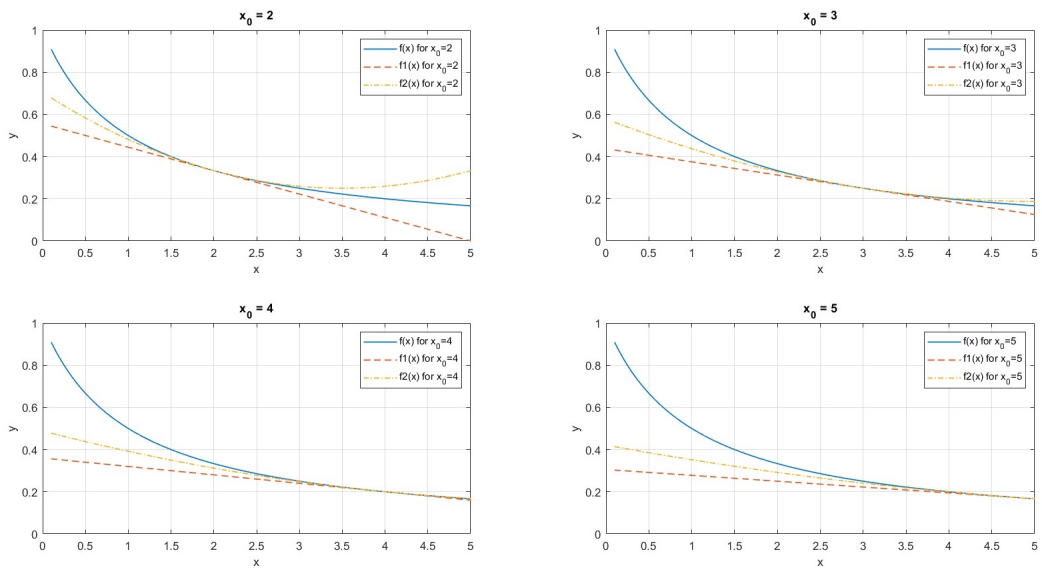

Figure 1: Function f and Taylor Approximations

We observe that as the value of x is increasing, the first and second order approximations are converging to f, after reaching the lowest point of the function.

2: Let f: ${\mathbb{R}_+}^2 \to \mathbb{R}$ with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$
(a) Compute and plot, via mesh, f for x_1,$x_2 \in [0, x_*]$,with $x_* > 0$.
(b) Plot the level sets of f, via contour. What do you observe? Can you explain the phenomenon?
(c) Compute the first- and second-order Taylor approximations of f at point $x_0 = (x_{01}, x_{02})$.
(d) Draw on a common plot, via mesh, f and its first-order Taylor approximation.
(e) Draw on a common plot, via mesh, f and its second-order Taylor approximation.
Answer: Using Matlab and the method "mesh" there was the following plot of function f. The code is given below.

Code:

```
% 1
% define function f(x1,x2)
f12 = @(x1, x2) 1./(1 + x1 + x2);

x1_range = 0:0.2:3;
x2_range = 0:0.2:3;

[x1,x2] = meshgrid(x1_range, x2_range);

f = f12(x1,x2);

figure;
mesh(x1,x2,f);
title('F(x) = 1/1+x1+x2')
xlabel('x')
ylabel('y')
zlabel('z')

grid on;
```

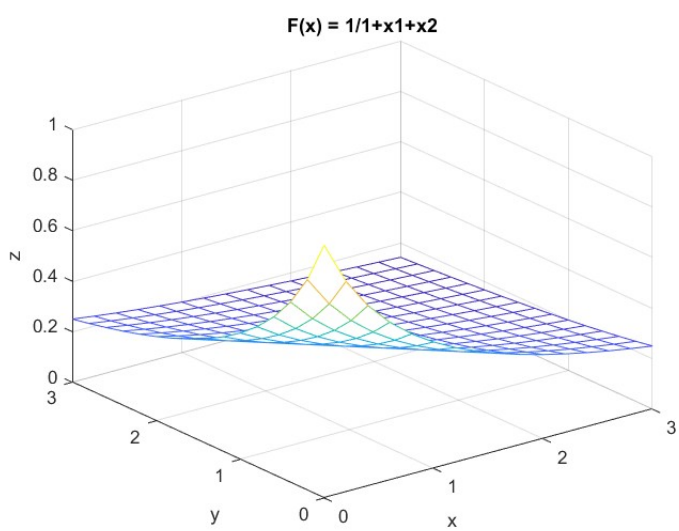

Figure 2: Function f

Following the same procedure for the question (b), there were extracted the folowing results.
Code:

```
% 2
contour_levels = linspace(0.1,0.9,10);

figure;
contour(x1,x2,f,contour_levels);
xlabel('x');
ylabel('y');
title('Level sets of f')
grid on;
```

By observing the level sets that came up, it is clear that they are convex. That happens because, as it is proved in exercise 5 below, the function f is actually, convex. The statement that sublevel sets of a convex function, are convex can be proven mathematically.

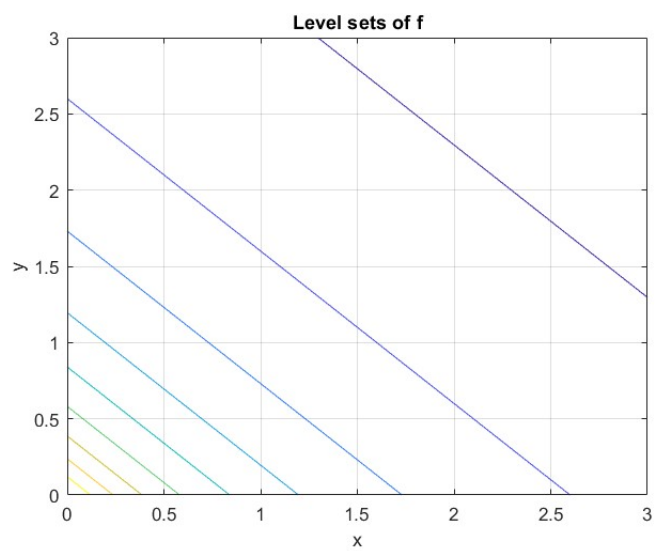

Figure 3: Level sets of function f

For the third part of the exercise we computed the first and second order Taylor approximations, by computing the partial derivatives and the Hessian matrix.

Code:

```
% 3
% we need the partial derivatives
part_x1 = @(x1, x2) -1/((1+x1+x2).^2);
part_x2 = @(x1, x2) -1/((1+x1+x2).^2);

part2_x1 = @(x1, x2) 2/((x1+x2+1).^3);
part2_x2 = @(x1, x2) 2/((x1+x2+1).^3);
part2_x1_x2 = @(x1, x2) 2/((x1+x2+1).^3);

x01 = 1;
x02 = 1;

x0 = [x01 ; x02];
x = [x1_range ; x2_range];

hf = [2/((x01+x02+1)^3),2/((x01+x02+1)^3) ; 2/((x01+x02+1)^3), 2/((x01+x02+1)^3)];

% first order approximation
f1 = f12(x01,x02) + part_x1(x01,x02) * (x1-x01) + part_x2(x01,x02) * (x2-x02);

% second order approximation
subs = x-x0;
f2 = f12(x01,x02) + ...
    part_x1(x01,x02) * (x1-x01) + part_x2(x01,x02) * (x2-x02) + ...
    (0.5).*(subs')*hf*(x-x0);
```

Then, we draw on a common plot the function and its approximations, separate plots for first and second approximation and we derived the following results.
Code:

```
% 4
figure;
mesh(x1_range, x2_range, f);
hold on;
mesh(x1_range, x2_range, f1,'EdgeColor','g');
xlabel('x1');
ylabel('x2');
zlabel('f');
title('Mesh Plot of f and Its First-Order Taylor Approximation');
legend('f(x1,x2)', 'First-Order Taylor Approximation');
hold off;

% 5
figure;
mesh(x1_range, x2_range, f);
hold on;
mesh(x1_range, x2_range, f2,'EdgeColor','r');
xlabel('x1');
ylabel('x2');
zlabel('f');
title('Mesh Plot of f and Its Second-Order Taylor Approximation');
legend('f(x1,x2)','Second-Order Taylor Approximation');
hold off;
```

The plots that are submitted are for $x_{0,1} = 1$ and $x_{0,2} = 1$.

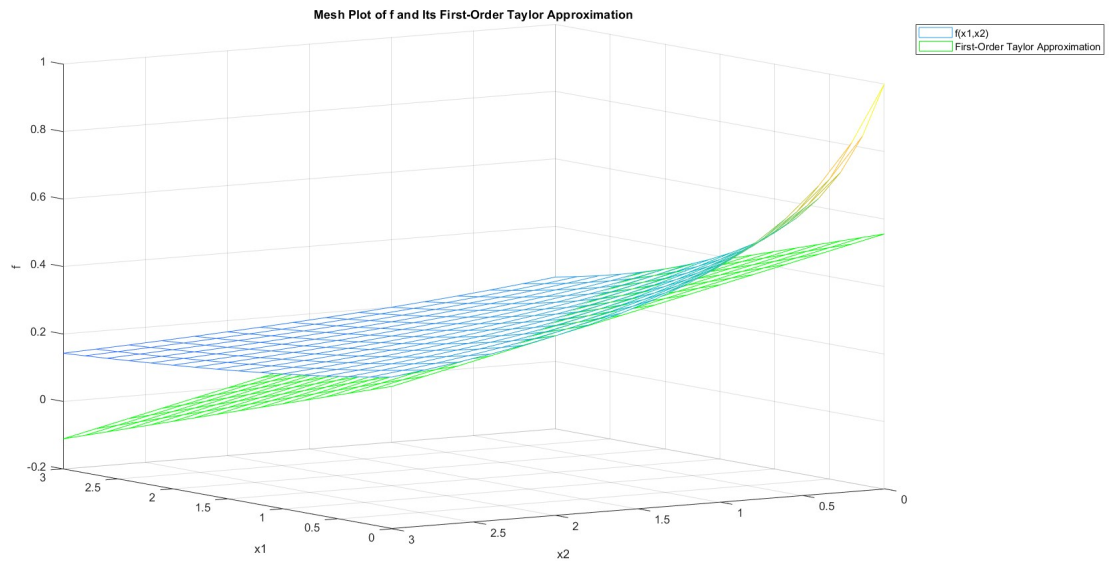

Figure 4: Function f and its First Order Taylor Approximation (green)

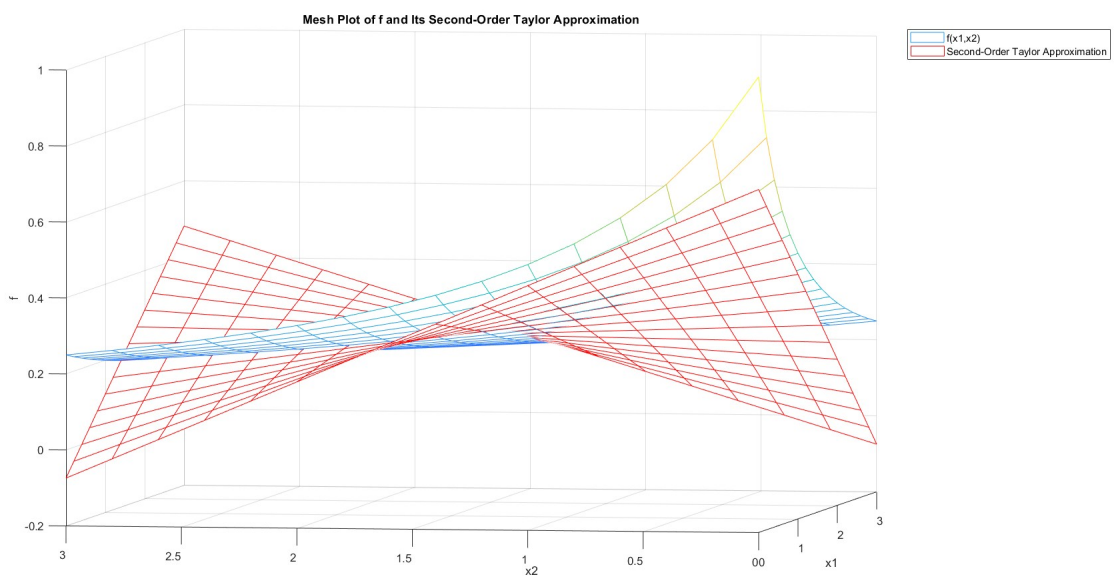

Figure 5: Function f and its Second Order Taylor Approximation (red)

3: Let $\mathbb{S}_{a,b} = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{a^T x} \leq b\}$.
(a) Prove that $\mathbb{S}_{a,b}$ is convex.
(b) Prove that $\mathbb{S}_{a,b}$ is not affine (a counterexample is sufficient).

Answer:
For the first part of the exercise the proof is the following: The convexity means that

$$\theta x_1 + (1-\theta)x_2 \in \mathbb{C}$$

Let $a^T x_1 \leq b$ and $a^T x_2 \leq b$.
So,

$$a^T(\theta x_1 + (1-\theta)x_2) = \theta a^T x_1 + (1-\theta)a^T x_2$$
$$\theta a^T x_1 + (1-\theta)a^T x_2 \leq \theta b + (1-\theta)b = b$$
$$\theta a^T x_1 + (1-\theta)a^T x_2 \leq b$$

So it is proven. For the second part we will give a counterexample. Let

$$a = (1,1) \;\; and \;\; b = 1$$
$$x_1 = (\frac{1}{2}, \frac{1}{2}), \;\; a^T x_1 = 1 \leq b$$
$$x_2 = (0,0), \;\; a^T x_1 = 0 \leq b$$

These are points that belong in the set $\mathbb{S}_{a,b}$. Let $\mathbb{S}_{a,b}$ be affine. Then all the linear combinations $x = \theta x_1 + (1-\theta)x_2$ have to satisfy the following relation $a^T x \leq b$. For $\theta = 2$ we observe that

$$a^T x = a^T(2x_1 - 0) = 2$$

So, $a^T x \leq b$ is not affine, but it is convex. That also leads to the conclusion that halfspaces, this is what $\mathbb{S}_{a,b}$ represents, are not affine.

4: Find the point $\mathbf{x}_*$ that is co-linear with $\mathbf{a}$ and lies on the hyperplane
$\mathbb{H}_{a,b} := \{x \in \mathbb{R}^n | a^T x = b\}$.
Answer:
The fact that $\mathbf{x}_*$ is co-linear is translated as $\mathbf{x}_* = \lambda a$ where $a \in R$. So,

$$a^T \lambda a = b \Rightarrow$$
$$\lambda = \frac{b}{a^T a}$$

So,

$$\mathbf{x}_* = \lambda a \Rightarrow$$
$$\mathbf{x}_* == \frac{b}{a^T a} a$$

5: Check whether the following functions are convex or not (using, for example, the second derivative rule).
(a) $f : \mathbb{R}_+ \to \mathbb{R}$ with $f(x) = \frac{1}{1+x}$
(b) $f : {\mathbb{R}_+}^2 \to \mathbb{R}$ with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$
(c) $f : \mathbb{R}_{++} \to \mathbb{R}$ with $f(x) = x^a$,for(to get a better feeling,plot function x^a for various values of a)
i. $a \geq 1$ and $a \leq 0$.
ii.$0 \leq a \leq 1$.
(d) $f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$ with $f_1(x) = \| x \|_2$ and $f_2(x) = \| x \|_2^2$ (plot the functions for $n = 2$).
Answer:
For question (a),

$$f(x) = \frac{1}{1+x}$$

$$\nabla f(x) = -\frac{1}{(1+x)^2}$$

and

$$\nabla^2 f(x) = \frac{2}{(1+x)^3}$$

We observe that $\nabla^2 f(x) > 0$ so f is a convex function for $x \geq 0$.
For question (b) we have,

$$f(x_1, x_2) = \frac{1}{1+x_1+x_2}$$

$$f'(x_1, x_2) = \nabla f(x_1, x_2) = (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \Rightarrow$$

$$f'(x_1, x_2) = (-\frac{1}{(1+x_1+x_2)^2}, -\frac{1}{(1+x_1+x_2)^2})$$

And for the second derivative we have

$$f''(x_1, x_2) = \nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

In this particular function we compute that

$$\frac{\partial^2 f}{\partial x_1^2} = \frac{\partial^2 f}{\partial x_2^2} = \frac{2}{(1+x_1+x_2)^3}$$

and

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = \frac{2}{(1+x_1+x_2)^3}$$

So the matrix is:

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{2}{(1+x_1+x_2)^3} & \frac{2}{(1+x_1+x_2)^3} \\ \frac{2}{(1+x_1+x_2)^3} & \frac{2}{(1+x_1+x_2)^3} \end{bmatrix}$$

In order to check if a function like f is convex, we have to check if the eigenvalues of f's Hessian matrix are non negative. The computed eigenvalues are $\lambda_1 = \frac{4}{(1+x_1+x_2)^3}$ and $\lambda_2 = 0$. These eigenvalues are non negative, because $x_1, x_2 \geq 0$ so the function is a convex one.

For question (c) we shall first examine the case where $a \geq 1$.

$$f'(x) = ax^{a-1}$$

and

$$f''(x) = a(a-1)x^{a-2}$$

$$a \geq 1$$

so $f''(x) \geq 0$ for that a. In case $a \leq 0$ we have that

$$a \leq 0 \;\; so \;\; a < 1 \Rightarrow a - 1 < 0$$

$$a(a-1) \geq 0$$

So, $f''(x) \geq 0$ convex. In general, f is convex for $a \geq 1$ and $a \leq 0$.
For the second part of the same question we examine the case where $0 \leq a \leq 1$ so,

$$f''(x) = a(a-1)x^{a-2}$$

$$a - 1 \leq 0$$

and

$$a \geq 0$$

so,

$$a(a-1) \leq 0$$

and due to that f is not convex for $0 \leq a \leq 1$. For a better feeling of the theoretical results, we designed the function x^a for various values of a. *Code:*

```
%% c
% first we plot for a >= 1 and a <= 0
x = linspace(0.1, 2, 5);
a_values_i = [1, -0.5, -1, 3, 0];
figure;
hold on;
```

```
for a = a_values_i
    y = x.^a;
    plot(x, y, 'LineWidth', 2, 'DisplayName', ['a = ', num2str(a)]);
end
hold off;
xlabel('x');
ylabel('f(x)');
title('f(x) = x^a for a \geq 1 and a \leq 0');
legend;
grid on;

% now for a>=0 and a <=1
a_values_ii = [0.5 , 0.8, 1];
figure;
hold on;
for a = a_values_ii
    y = x.^a;
    plot(x, y, 'LineWidth', 2, 'DisplayName', ['a = ', num2str(a)]);
end
hold off;
xlabel('x');
ylabel('f(x)');
title('f(x) = x^a for 0 \leq a \leq 1');
legend;
grid on
```

We received the following results:

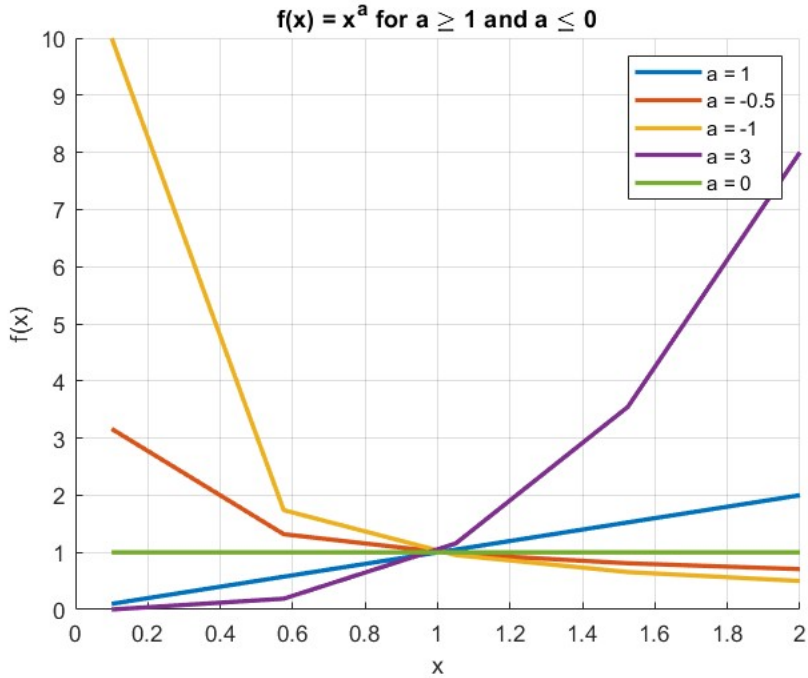

Figure 6: $f(x) = x^a, for \;\; a = -1, -0.5, 0, 1, 3$

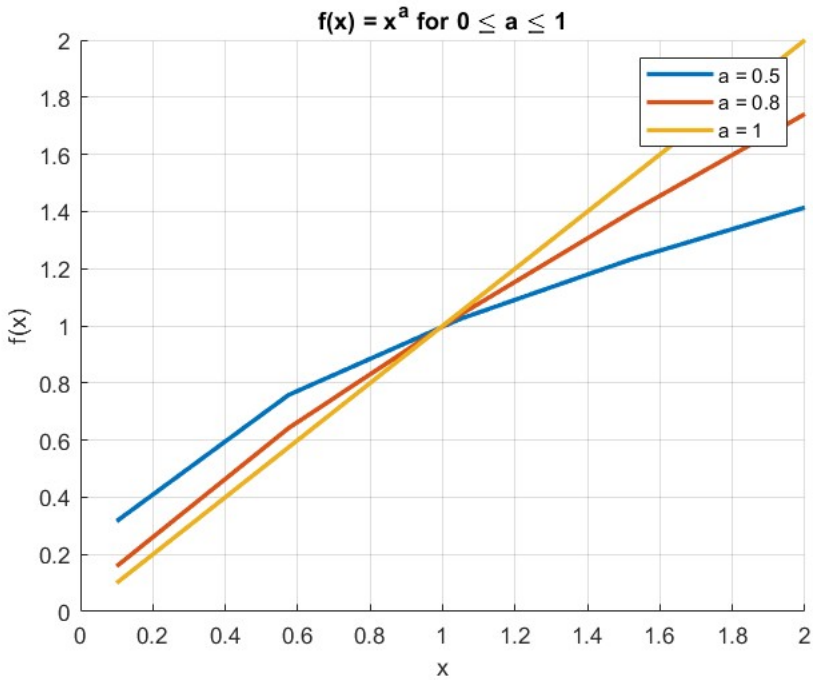

Figure 7: $f(x) = x^a, for \;\; a = 0.5, 0.8, 1$

For question (d) we will use the definition of convexity. We will calculate the Euclidean norm of x and y where $\theta x + (1-\theta)y$, where $x, y \in R$.

$$f_1(\theta x + (1-\theta)y) = \| \theta x + (1-\theta)y \|_2$$

We will use the Cauchy-Schwarz inequality where $\| a + b \|_2 \;\leq\; \| a \|_2 + \| b \|_2$. So we have that

$$\| (\theta x + (1-\theta)y) \|_2 \;\leq \theta \| x \|_2 + (1-\theta) \| y \|_2 \Rightarrow$$

$$f_1(\theta x + (1-\theta)y) \leq \theta f_1(x) + (1-\theta) f_1(y)$$

By that we conclude that the Euclidean norm of $f_1(x) = \| x \|_2$ is convex.
Then, we move to the $f_2(x) = \| x \|_2^2$. We have that

$$f_2(\theta x + (1-\theta)y) = \| \theta x + (1-\theta)y \|_2^2$$

and following the Cauchy-Schwarz inequality , as before, we have that

$$\| \theta x + (1-\theta)y \|_2^2 \;\leq\; \theta \| x \|_2^2 + (1-\theta) \| y \|_2^2 \Rightarrow$$

$$f_2(\theta x + (1-\theta)y) \;\; \leq \theta f_2(x) + (1-\theta) f_2(y)$$

Then, we proceeded to plot the the functions for n=2. *Code:*

```
x1_range = -2:0.1:2;
x2_range = -2:0.1:2;

[x1,x2] = meshgrid(x1_range, x2_range);
f1 = sqrt(x1.^2 + x2.^2);
f2 = f1.^2;

figure;
mesh(x1,x2,f1)
title('Euclidean Norm f1');

figure
mesh(x1,x2,f2)
title('Square of Euclidean Norm')

xlabel('x1')
ylabel('x2')
```

The results we got:

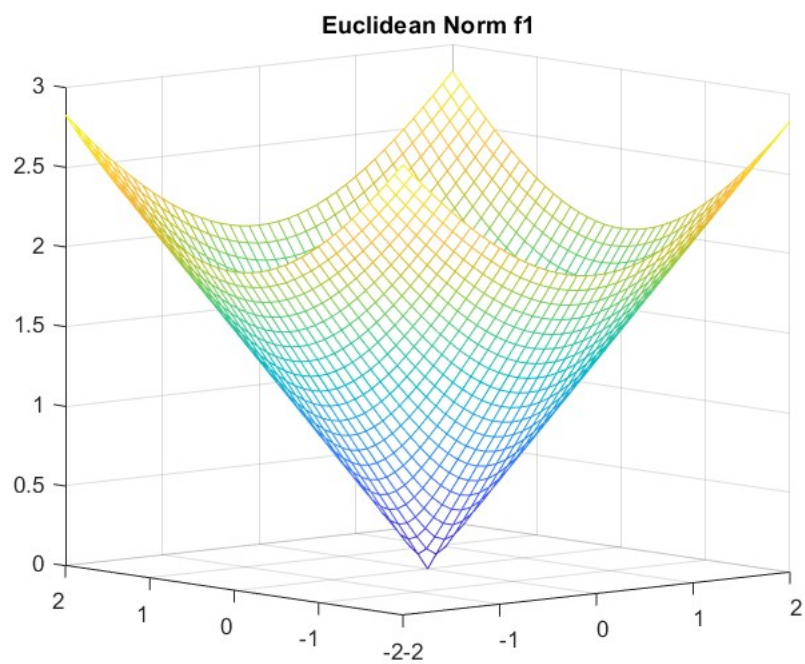

Figure 8: The Euclidean Norm, function f1

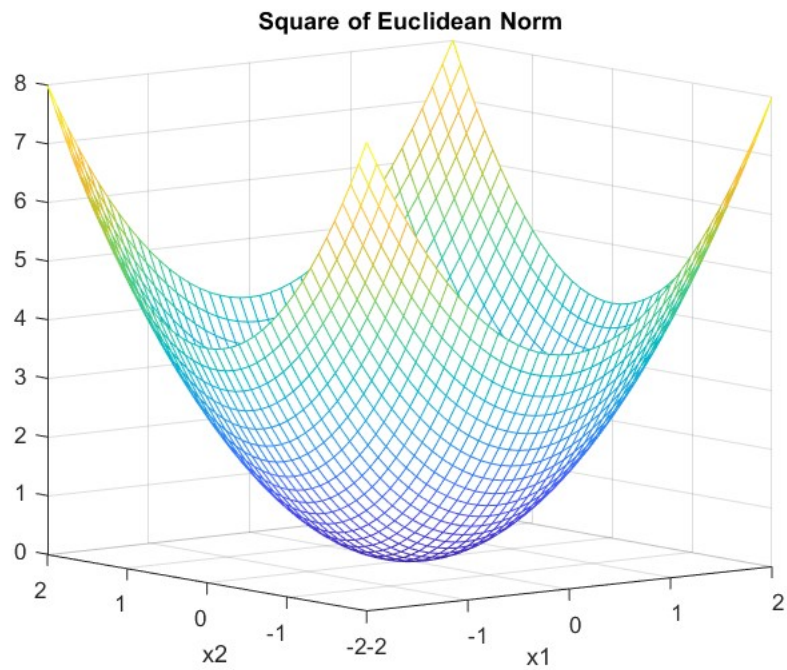

Figure 9: Function f2

6: In the sequel, we shall study the very important convex quadratic functions. Let $\mathbf{P} \in \mathbb{S}^2_{++}$ where $\mathbb{S}^2_{++}$ in the set of the (2×2) positive define matrices $\mathbf{q} \in \mathbb{R}^2$, and $r \in \mathbb{R}$.Consider the convex quadratic function f: $\mathbb{R}^2 \to \mathbb{R}$,define as:

$$f(x) = \frac{1}{2}\mathbf{x^T P x} + \mathbf{q^T x} + r$$

It can be shown (we shall prove it later) that the function f is minimized at point

$$x_* = -\mathbf{P^{-1}q}$$

a) Plot function f, via mesh and contour around $\mathbf{x}_*$, and verify, visually, the claim about the optimal point $\mathbf{x}_*$.

Answer:
Using the directions that were given, function f and the optimal point were designed. The results that were extracted are presented below:

```
P =

    4.6094    1.2399
    1.2399    0.5584


q =

     2
     3


x_optimal =

   -0.9953   -0.3196
```

Figure 10: Matrix P, q and the optimal point x

Due to the use of the Matlab method randn(), every time that the code is run there will be a different result.

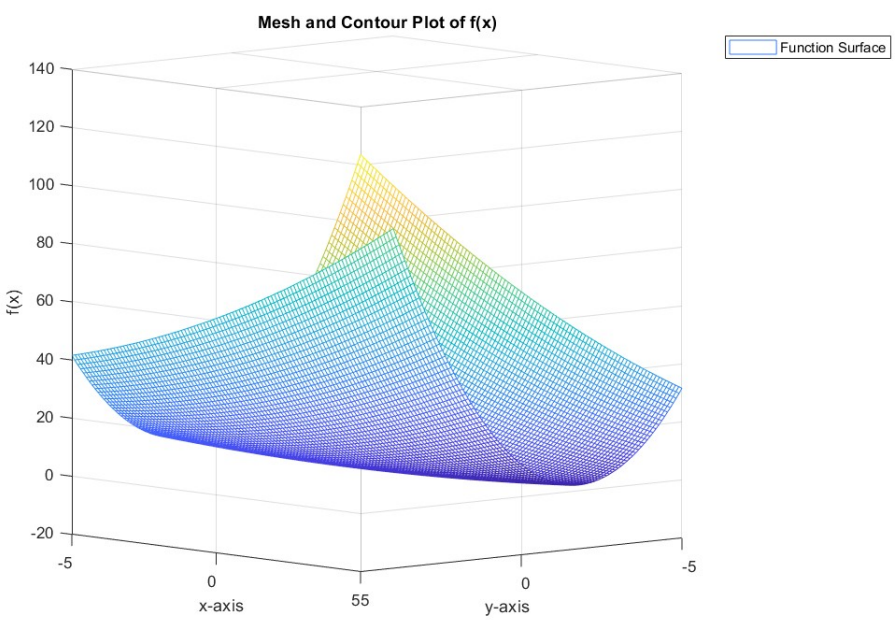

Figure 11: Mesh of funtion f

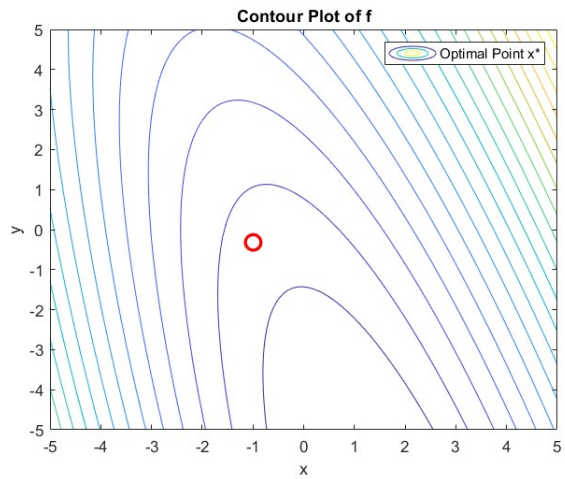

Figure 12: Optimal point

Code:

```
%% Sixth Exercise
% first, we generate the P as indicated
A = randn(2,2);
P = A*A';

q = [2; 3];
r = 3;

% we define the range for the coordinates x and y
x1_range = -5:0.1:5;
x2_range = -5:0.1:5;

[x1,x2] = meshgrid(x1_range, x2_range);

f= 0.5 * (x1.^2 * P(1, 1) + x2.^2 * P(2, 2) + ...
    (x1 .* x2) * (P(1, 2) + ...
    P(2, 1))) + q(1) * x1 + q(2) * x2 + r;
P
q
x_optimal = -q\P

figure;
mesh(x1, x2, f);
```

```
xlabel('x-axis');
ylabel('y-axis');
zlabel('f(x)');

title('Mesh and Contour Plot of f(x)');
legend('Function Surface', 'Contour Lines', 'Optimal Point');
grid on;

figure;
contour(x1, x2, f, 20);
hold on;
plot(x_optimal(1), x_optimal(2), 'ro', 'MarkerSize', 10, 'LineWidth', 2);
xlabel('x');
ylabel('y');
title('Contour Plot of f');
legend('Optimal Point x*');
```